

AIR DATA BASICS

FROM AN FTE PERSPECTIVE

Darren G. McDonald^F Technical Fellow, Flight Test Engineering
Telephone: 206-473-2310 E-mail: darren.g.mcdonald@boeing.com

*The Boeing Company
Seattle, WA 98124*

As part of introductory aerodynamics, several different variations of airspeed and altitude are introduced, and their derivations using concepts such as the hydrostatic equation and the isentropic equations are explained in detail. The practical application of airspeed and altitude during flight test of air vehicles, and how to visualize and understand the relationships between each, is often not addressed - leading to confusion, incorrectly flown test points, and sometimes even exceeding limits. This paper presents each of the different airspeeds in a progressive fashion, addresses the basic different types of altitude, and then provides some easy methods to remember how they all relate to one another. Finally, this paper illustrates the magic behind pressure altitude, and when and why it is such a useful tool in flight test.

I. INTRODUCTION

Flight test is an exacting and unforgiving endeavor, with operating envelopes and parameters that are often the deciding factor between success and failure. One of the most important envelopes of any air vehicle while airborne is the flight envelope – the combination of altitude and airspeed within which the aircraft can be safely flown. Many flight envelopes include limits that are defined as a function of several different types of airspeeds, and there are numerous ways in which altitude can be measured and displayed. For a flight test engineer, it is critical to understand the source of these different measurements and how they relate to one another. Further, by establishing memorable ways to conceptualize these measurements and understand the physics at their source, selecting the best airspeed and altitude for each scenario becomes much simpler. The flight test engineer must design test points by selecting airspeed and altitude parameters that are available to the pilot, while fulfilling the needs of the aircraft design engineers. Sometimes the requirements are defined as terms that are not displayed to the flight crew, so the flight test engineer must utilize equivalent parameters to ensure the test is flown correctly. Doing so successfully requires an understanding of how the airspeed and altitude measurements inter-relate. Finally, it is important to note that the scope of this paper is limited to the aspects of air data that affect the planning and execution of flight test, and as such it does not cover density or geopotential altitudes, nor non-dimensionalization or expansion of performance data results.

II. AIRSPEED – YOU CAN’T PICK ONE

Airspeed measurement seems simple enough, you need a pitot and a static source, some basic tubing, and a mechanical device that can display airspeed. Aerodynamics textbooks are filled with derivations of equations and theory, however, and very quickly one finds that there are several different airspeeds that all have simplifications and assumptions attached. These differences mean that each form of airspeed has separate applications for which their use is necessary. The most common forms that airspeed can take are calibrated, indicated, equivalent, and true. It is vitally important for a Flight Test Engineer to understand the relationships between each, how and when to apply each correctly, and how the variations in atmospheric conditions affect their relationships.

A. Pressures – Total, Static, Impact, and Dynamic

Total pressure, P_T , is the pressure that is measured when the free-stream velocity is brought to zero velocity. Any stagnation point will result in total pressure being measured, which is the basic principle used by Henri Pitot’s invention of the pitot tube in the early 1700s.

Static pressure, P_s , is the atmospheric pressure that is measured by the static port. When there is no airflow, static pressure should be equal to ambient pressure, P_a . Measurement of static pressure on a moving aircraft always requires a correction to more accurately represent ambient pressure.

Impact pressure, q_c , is defined as the difference between total pressure and corrected static pressure. It must be noted that impact pressure is affected by compressibility effects. The compressibility effects become noticeable above approximately 200 knots and should be accounted for at speeds greater than 250 knots.

$$q_c = P_T - P_s \quad \text{eq. 1}$$

Dynamic pressure, $\bar{q}$ (qbar), is a measure of the kinetic energy of an airflow, and compressibility effects are accounted for.

$$\bar{q} = \frac{1}{2} \rho V^2 \quad \text{eq. 2}$$

where ρ is the density of the air, and V is airspeed. Note that the subscript of density and velocity must be congruent, as will be addressed later in this paper.

A common childhood aerodynamic experiment is to stick a hand out an open car window and begin ‘flying’ an outstretched palm. The forces generated scale as the speed of the car increases with the square of the velocity, as shown in equation 2. The maximum force generated is with the palm perpendicular to the airstream, creating the largest possible stagnation area, and vividly linking impact pressure to bruises on the child’s forearm due to impact with the aft window frame. At automobile speeds and altitudes the impact pressure and dynamic pressure will be virtually indistinguishable values.

B. It all starts with Bernoulli

The basis of the computations for airspeed is Bernoulli's equation for compressible flow.

$$\frac{\gamma}{(\gamma-1)} \frac{P_s}{\rho_s} + \frac{1}{2} V^2 = \frac{\gamma}{(\gamma-1)} \frac{P_T}{\rho_T} = \text{constant} \quad \text{eq. 3}$$

where γ is the ratio of specific heats, which is a constant of 1.4 for air, P_s is the static air pressure, ρ_s is the static air density, V is the air velocity, P_T is the total pressure, and ρ_T is the total density.

The various forms of this equation that are used to compute each of the different types of airspeed (including Mach) are referred to as the isentropic equations, such as presented in equation 4. This subject is well covered in Sections 3 and 4 of the SFTE Reference Handbook, plus various aerodynamic and flight test textbooks, so will not be repeated here. Further, the intent of this paper is to focus on the relationships between each, so instead of calculating each directly from the isentropic form, the mathematical relationships between each will be presented. This is not recommended in general practice, but is useful for visualization and understanding.

C. Calibrated (KCAS or V_c) Airspeed

Calibrated airspeed is the airspeed that is displayed to the flight crew for the majority of modern aircraft (with digital flight instruments). It is often noted as KCAS, or knots calibrated airspeed, as knots is the predominant standard in aviation.

If we substitute a pitot tube for the perpendicular palm of our childhood hand-flying days, and measure the static pressure perpendicular to the airflow, we can then apply the following rearranged version of equation 3 as follows:

$$V_c = KCAS = 1479.1 \sqrt{\left[\left(\frac{P_T - P_s}{P_o} + 1 \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} \quad \text{eq. 4}$$

where γ is the ratio of specific heats, P_o is the ambient pressure at sea-level on a standard day, P_T is the total pressure, and P_s is the corrected static pressure. The constant in front of the equation includes unit conversion as well as several constants, which will be discussed in more detail below.

From equation 4 above, we can see that the constant for sea-level standard day ambient pressure is embedded. In addition, the numerical constant 1479.1 contains several terms, which are as follows:

$$1479.1 = \frac{\sqrt{\frac{2\gamma P_o}{(\gamma-1)\rho_o}}}{1.6878} \quad \text{eq. 5}$$

where 1.6878 is the conversion from knots to feet per second, and ρ_o is the sea-level standard day density.

The constant from the calibrated airspeed equation also has sea-level standard day static pressure and density, as well as the ratio of specific heats for air, in addition to a unit conversion factor. It is highly unlikely, if not virtually impossible, for an aircraft to be flying at perfect standard day atmospheric ambient pressure and density conditions! Therefore, the *calibrated airspeed* represents an airspeed with both a pressure ratio (δ) and density ratio (σ) error as compared to the true airspeed.

D. Indicated (KIAS or V_i) Airspeed

Indicated airspeed is simply the airspeed that is calculated and displayed if no corrections are applied to total or static pressure measurement inaccuracies. For fixed-wing aircraft in normal flight, total pressure error is nominally all due to instrument error, with no significant position error. Static pressure error (ΔP_s) is commonly referred to as installation error or position error, and can be also denoted as an airspeed error, ΔV_p . Therefore, equation 4 will result in indicated airspeed if the static pressure has not had the position error removed. If the position error is available in terms of airspeed, then the following simple equation may be applied:

$$V_c = V_i + \Delta V_{ic} + \Delta V_{pc} \quad \text{eq. 6}$$

where V_c is the calibrated airspeed, V_i is indicated airspeed, ΔV_{ic} is the instrument correction for both static and total pressure error, and ΔV_{pc} is the airspeed correction due to position error (usually only applies to static pressure).

E. Equivalent Airspeed (KEAS or V_e)

If we go back to our childhood aerodynamic experiments, we'll ride along on a family vacation where dad hauls the family station wagon to the top of Pike's Peak, Colorado at an elevation of 14,110 feet. At this altitude, the static pressure will have dropped by over 40% since leaving sea-level, and the air density will have dropped over 35%. For a constant ground speed, the static pressure and total pressure will have both decreased, although not in equal proportions, resulting in a decrease in impact pressure as well. If the family wagon had a calibrated airspeed indicator, and there wasn't any wind, the trip would result in a reduction of "how much longer" calls from the back, as the ground speed would keep increasing with elevation to maintain a constant impact pressure. We can cut even more of our precious travel time by switching to the more accurate equivalent airspeed - although mom's nerves and the highway patrol would be liable to have a particularly negative perspective of doing so! Equivalent airspeed removes the pressure ratio error from calibrated airspeed as follows.

When you are not restricted by mechanical devices, and if you have sufficient measurements available, it is pretty straightforward to take the actual corrected static pressure and replace all of the instances of sea-level static pressure in equations 4 and 5 with the corrected static pressure to obtain the equivalent airspeed, per the equation below.

$$V_e = KEAS = \sqrt{\frac{2\gamma P_s}{(\gamma-1)\rho_o} \left[\left(\frac{P_T - P_s}{P_s} + 1 \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} \quad \text{eq. 7}$$

where the constant has been replaced with γ (ratio of specific heats), P_s (corrected static pressure), and ρ_o (sea-level standard day density).

The equation to convert from calibrated airspeed to equivalent airspeed is not simple, but by using the pressure ratio, δ , which is P_a/P_o (local ambient pressure, P_a , divided by sea-level standard day ambient pressure, P_o), the following conversion equation can be used:

$$V_e = KEAS = 1479.1 \sqrt{\delta \left[\left(\frac{1}{\delta} \right) \left\{ 1 + 0.2 \left(\frac{V_c}{661.4786} \right)^2 \right\}^{\frac{7}{2}} - 1 \right] + 1}^{\frac{2}{7}} - 1 \quad \text{eq. 8}$$

The relationship between the two as a function of pressure altitude (shown as pressure ratio in equation 8 above) is therefore deterministic, so the correction can also be looked up from charts that are readily available, such as in Section 4.2 of the SFTE Reference Handbook.

The correction to equivalent airspeed is often referred to as a compressibility correction, although this is somewhat misleading – recall that we started with Bernoulli’s equation for compressible flow! It is primarily a correction for local ambient pressure from sea-level ambient pressure, or a removal of the airspeed error due to the P_o constant utilized in the calibrated airspeed equation.

Equivalent airspeed is extremely important for aerodynamics as well as structural load. A constant equivalent airspeed is a constant dynamic pressure, and represents a constant air load on the aircraft. As a matter of fact, if we look at the equation for dynamic pressure (equation 2 above), and look ahead to the equation for the relationship between true and equivalent airspeed, the equation for dynamic pressure is below:

$$\bar{q} = \frac{1}{2} \rho_o V_e^2 \quad \text{eq. 9}$$

An additional note here is that all structural loads related measurements, such as side-load due to sideslip angle, must use dynamic pressure in their calculations, *NOT* impact pressure, which will underestimate the total load. Also, recall that dynamic pressure is a dominant term in the calculation of lift, drag, and pitching moment coefficients, and therefore equivalent airspeed is important as the primary variable to assess lift, drag, and pitching moment behavior as a function of altitude (until the aircraft accelerates into regions associated with shockwave interactions due to increasing Mach). If an aircraft climbs in altitude at a constant equivalent airspeed, the calibrated airspeed will increase in a nonlinear fashion.

F. True Airspeed (KTAS or V_T)

A final correction to equivalent airspeed must be made in order to reach the most correct possible airspeed for navigational purposes. Recall the equations for equivalent airspeed all retain a constant for sea-level standard day ambient density. As our local air density changes as a function of atmospheric conditions and altitude, this introduces error.

If we return back to our childhood analogies, in an admittedly somewhat strained but memorable example, we can look at kite flying. This is the first experience where we learn about the difference between groundspeed and airspeed. Invariably, the available windspeed is not great enough to keep the kite flying, so we must resort to running to add to the apparent airspeed of the kite in the hopes of creating enough lift for the kite to remain airborne. The combination of groundspeed and windspeed results in the airspeed velocity of the kite, which either stays aloft or crashes to the ground. If some type of windspeed measurement device was available, an adult could have done some quick calculations to determine if we could run fast enough to make up the windspeed deficiency. Windspeed will always be measured in true airspeed, so whenever wind calculations are required the aircraft velocity must be in true airspeed. It is helpful to remember this experience as a method to associate wind speed and also speed of sound to true airspeed. We can also make use of this scenario as a reminder that if we subtract our groundspeed vector from our true airspeed vector we can calculate our windspeed vector. Dad learned this the hard way using his airspeed indicator driving home from family vacation in Colorado with a big tailwind – the police were not amused with his explanation of why he was driving well over the speed limit! Thus to be able to reliably navigate, or calculate our wind conditions, we must first determine how to calculate our true airspeed.

If we already have our equivalent airspeed, we simply need to correct for the local air density. The air density ratio, σ , is ρ_a/ρ_o , or local ambient air density (ρ_a) divided by sea level standard day ambient density (ρ_o). As a result, the equation for true airspeed (V_T) from equivalent airspeed is as follows:

$$V_T = \frac{V_e}{\sqrt{\sigma}} \quad \text{eq. 10}$$

There is only one airspeed left, and that is the Mach number, or the speed of the aircraft calculated as a ratio to the speed of sound. Mach number is a very simple equation:

$$M = \frac{V_T}{a_o \sqrt{\theta}} \quad \text{eq. 11}$$

where a_o is the speed of sound at sea-level on a standard day, and the temperature ratio, θ , is T_a / T_o , with T_a the ambient (static) temperature at the current altitude and T_o is the sea-level standard day ambient temperature.

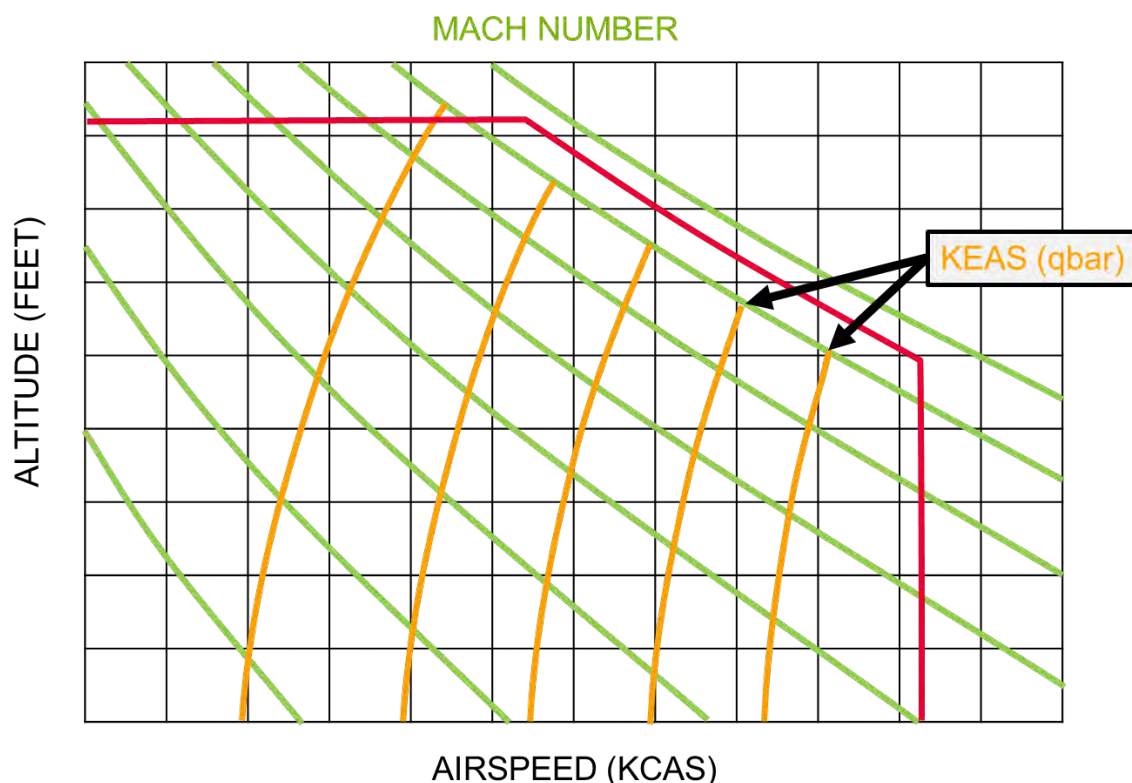
By re-arranging the isentropic equation you can also compute Mach directly from the equivalent airspeed using only the pressure ratio – no temperature required!

$$M = \frac{V_e}{661.4786} \sqrt{\frac{1}{\delta}} \quad \text{eq. 12}$$

where V_e must be in knots for the 661.4786 conversion factor

The reason we are able to do this is that the pressure ratio, δ , is equal to the density ratio, σ , multiplied by the temperature ratio, θ . $\delta = \sigma * \theta$

Of all the speeds covered so far, we will soon see that we can plan for and predict the relationships between all but one very predictably and easily. For a very effective reference as both a memory aid and as a visualization of the relationships, a flight envelope from any aircraft may be used. Below is a generic example of an aircraft flight envelope. The black grid illustrates horizontal lines of constant altitude and vertical lines of constant calibrated airspeed. The orange lines are lines of both constant equivalent airspeed as well as lines of constant dynamic pressure. Finally, the green lines are lines of constant Mach, and the red lines are the dive and altitude limits. Note that true airspeed is not included. One may also ponder what happens to these parameters on a non-standard atmospheric day? To understand why true airspeed is missing, and what the effects of a non-standard day on the flight envelope are, we must look at altitude. In fact, the keen observer may notice a deliberate omission in the following flight envelope.



III. ALTITUDES – WHICH ONE IS ‘RIGHT’?

In a similar manner to airspeed, there are numerous ways to describe altitude, and there is often confusion due to the variety and subtleties between each. Thankfully, there are just two main categories of altitude: geometric and pressure-sensed. Each category of altitude contains a variety of altitudes, and knowing which altitude is available - and how to apply it - is critical to a successful flight test.

A. Tapeline Altitudes

A tapeline altitude is one which could be measured with a very long tape measure from the reference point on the aircraft to the ground reference point. Perhaps the most commonly used form of tapeline altitude is geometric altitude (h_G), which uses a ground reference point of sea level. When conducting terminal area operations, or when clearance to obstacles is a concern, geometric altitude is conceptually the best answer, as most aviation charts depict obstacles and airport facilities in terms of mean sea level (msl). Arguably the biggest challenge with geometric altitude is making sure that you understand the zero datum, and know how to make use of that information. For example, GPS altitude is all based on the ellipsoid model (usually the WGS 84), so unless your system reports local geoid height, the GPS altitude will be ellipsoid height, differing by up to several hundred feet from your height above ground level. The altitude difference between the ellipsoid and geoid heights is a function of your latitude and longitude, so can be easily converted as needed. The geoid height is also considered as the mean sea-level (MSL) height for aviation purposes.

Radio altimeters will just report the distance above ground level (agl), or any solid objects built on it. Any elevation changes in the terrain below the aircraft while it is flying in level flight will show as a change in radio altimeter, or tapeline distance to the ground, which is a moving datum.

B. Pressure-based Altitudes

The simplest and most robust method of measuring altitude is one that is based on sensing the local pressure. The corrected static pressure measurement can be calibrated and displayed in terms of altitude. Depending on whether you are in the troposphere (> 6.683 inches of mercury), in the tropopause, or higher, the appropriate equation uses the pressure ratio to calculate the pressure altitude. There is a significant caveat that comes with pressure altitude – it only provides valid geometric altitude readings on a standard atmospheric day! If there is an atmospheric high or low pressure system, or if the standard day pressure lapse rate is above or below the ISA standard atmosphere model, then the pressure altitude and the geometric altitude will disagree. Experience proves that a perfect standard atmospheric day almost never exists. Worse yet, as you traverse from one weather system to the next your pressure altitude will vary. For example, if you fly from a high pressure system towards a low pressure system your geometric altitude will decrease while maintaining a constant pressure altitude – thus the common private pilot training memory aide of “From high to low, look out below!”.

There are three different altimeter settings that are commonly applied to pressure-based altitude, each of which must be defined and understood to ensure safe and efficient flight test execution. The three altimeter settings are QNH, QFE, and QNE. There is no literal meaning to these confusing letters, they were simply developed in the early 1900s for ease of transmitting information via Morse code. These abbreviations are also used in industries beyond aviation.

1. QNH altimeter setting

This is the most commonly used altimeter setting for flying in and around airports. When QNH is set, the aircraft altimeter will read the airport elevation above mean sea level. For departure and approach segments, the altimeter will display the aircraft height above mean sea-level. Navigation

charts are published with obstacle heights in mean sea-level, so using QNH provides the simplest and most intuitive altitude for aircrews concerned about maintaining departure and approach altitude profiles. The QNH setting is provided via ATIS, or sometimes directly from the air traffic controller. What if QNH is unavailable, or if there is any question about the proper QNH setting? If the aircraft is parked at a known field elevation, the altimeter setting window can be adjusted until the displayed altitude on the altimeter matches the known height above mean sea-level.

2. QFE altimeter setting

In some areas of the world, the standard used is QFE. When the altimeter setting is QFE, the altimeter will read zero when the aircraft is on the ground at the airport where the reading was provided. In this case, during the departure and approach phases of flight the altimeter will display the height of the aircraft above the field elevation. This method of altimeter setting is used as a cheap and effective radio altimeter, most often used in this manner by aerobatics pilots. In a flight test environment, QFE can be set by adjusting the altimeter to read zero when on the ground, or by adjusting the altimeter setting to match the current radio altitude, if available.

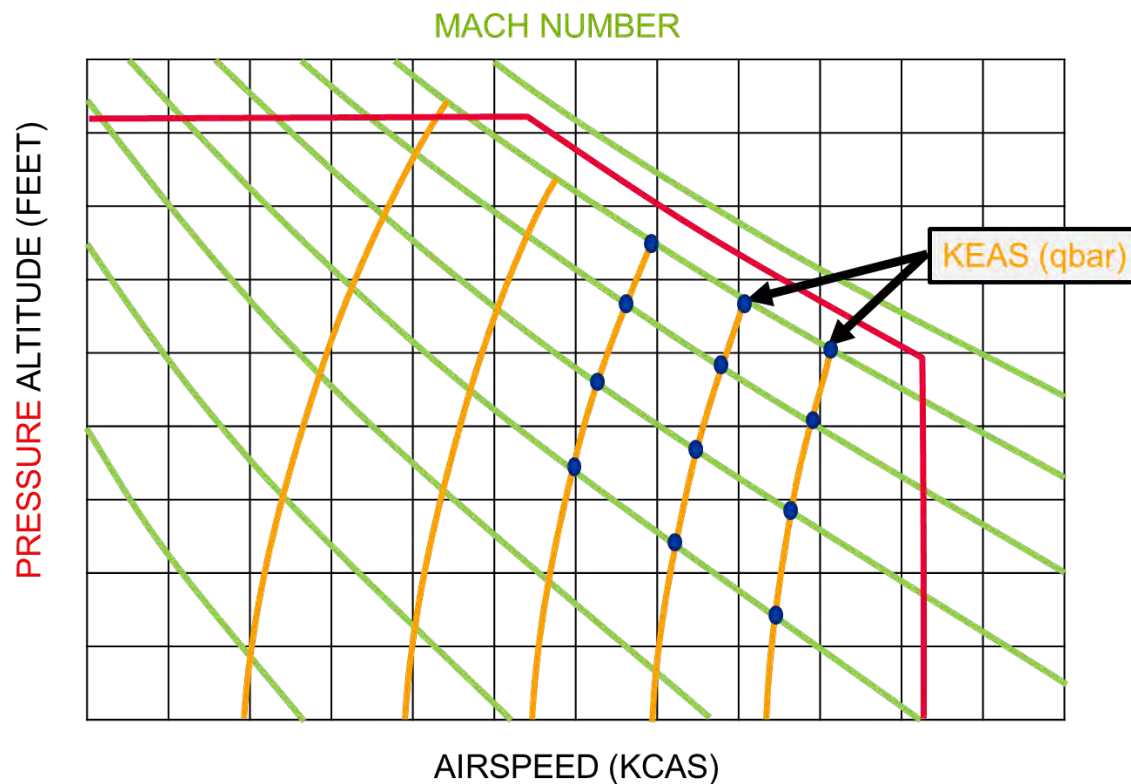
3. QNE altimeter setting (H_p , or pressure altitude)

The final altimeter setting, QNE, should look familiar as it is the standard day atmospheric pressure setting – 29.92 inches of mercury, or 1013.2 hectoPascals. The QNE altimeter setting is effectively no altimeter setting correction at all, and results in the altimeter displaying the standard day pressure altitude. During the average transcontinental flight, it is common for several pressure systems to be traversed. If the pilots of jet aircraft were tasked with keeping their QNH or QFE altimeter settings accurate and up to date during cruise, the end result would be clogged radio transmissions and human errors leading to disastrous outcomes. To avoid this issue, and to provide a robust solution to keeping aircraft that are at altitudes well above ground obstacles safely separated from one another, the use of transition altitudes was implemented. A transition altitude is defined by each aviation regulatory body, with the United States and Canada observing a transition altitude of 18,000 feet. Above the transition altitude, an altimeter setting of QNE must be used. To distinguish the altimeter setting above the transition altitude from the altitudes below it, the term flight level (FL) is used. A flight level is pressure altitude (QNE) divided by 100, so FL320 is 32,000 feet pressure altitude.

It should be noted that temperature variations also affect pressure-based altitudes and their relationship to geometric altitudes, although the author has elected to leave those details out of the discussion as beyond the scope of this paper – see Sections 3.3 and 3.4 of the SFTE Reference Handbook for discussion on geopotential, density, and temperature altitudes.

Now that we have covered altitudes, it is time to look at our generic flight envelope once again. As alluded to earlier, the original figure presented in this paper above was missing one key qualifier. Recall the common parameter in each of the items involved in the flight envelope. Starting with calibrated airspeed, the only additional parameter needed to convert to equivalent airspeed, dynamic pressure, or Mach is the pressure ratio. Pressure altitude is the pressure ratio displayed in units of feet, with no altimeter setting adjustment. Note that true airspeed is a function of temperature ratio, and therefore it cannot be included on the flight envelope along with altitude and other airspeeds, as the temperature ratio has no dependency on pressure

altitude. It follows directly then, that pressure altitude can be plotted on a y-axis and the related airspeed terms can be plotted as a visual representation of their relationships. The blue dots added to the figure represent a common test matrix technique to minimize points while capturing Mach and dynamic pressure effects. It is relatively straightforward to establish the test matrix as a function of pressure altitude, and then to specify either Mach or KCAS as appropriate to provide an operationally simple and robust way to ensure the desired data are gathered. The label of Pressure altitude has been added in red in the figure below to help remember how important that qualifier is. Further, referring to a flight envelope is a simple and effective way to recall which airspeeds are related to pressure ratio, and how they are related to each other as a function of pressure altitude.



IV. CONCLUSION

The main terms related to airspeed and altitude measurement and display in aviation have relationships that must be understood to ensure safe and efficient flight test execution. The terms can be made more easily memorable and visualized with a few simple techniques. The easiest and most complete way to do so is by using a flight envelope such as those published with a typical commercial transport aircraft. Find a method that helps you make the terms memorable – the author has used the following, some of which are tongue in cheek, to make these terms more intuitive and recallable.

Static is not moving, so **static pressure** is the pressure you read when there is *no relative wind*. **Total pressure** is the complete, or *highest, air pressure* you can measure.

Impact pressure is what makes your arm impact the window frame in the car – it is *total pressure minus static pressure*.

Dynamic pressure is related to the square of the velocity, and it is a true representation of the *dynamic forces acting on an aircraft (such as drag, lift, and pitching moments)*.

Calibrated airspeed is a *calibrated lie* – it is the airspeed with a sea-level pressure error and a sea-level density error.

Indicated airspeed is a *sugar-coated calibrated lie* – instrument and position errors included.

Equivalent airspeed is a constant, or *equivalent, structural load, lift, drag, and pitching moments at all altitudes*. Equivalent airspeed still has a sea-level density error.

Mach sounds funny, and it is related to the *speed of sound*. Oddly enough, Mach can be calculated as a function of equivalent airspeed and pressure ratio, so it can be plotted as a function of pressure altitude!

True airspeed is the airspeed that *tells no lies* – you can subtract your groundspeed vector and determine your wind vector. The trouble with true airspeed is that it requires the density ratio, which is a function of temperature, and is therefore not accounted for by pressure altitude. You will not find true airspeed on a flight envelope alongside pressure altitude and other airspeeds!

One of the primary responsibilities of a flight test engineer is to operationalize design engineering test requirements in a safe and efficient manner, to ensure the maneuvers are flown correctly, and to ensure the correct data are gathered. This makes pressure altitude a flight test engineer's best friend. Whether a test requirement is related to a dynamic pressure, a test matrix of constant Mach and dynamic pressure lines, or a calibrated or equivalent airspeed limit, specifying the test condition as a function of pressure altitude unlocks critical capability. If pressure altitude is utilized, the flight test engineer can select Mach or calibrated airspeed (often the only speeds displayed on production instruments), and confidently predict the equivalent airspeed, dynamic pressure, and the Mach or calibrated airspeed that was left unspecified. *The result is guaranteed regardless of atmospheric variations from standard day!*

The only remaining responsibility for the flight test engineer is to monitor the altimeter setting during test execution. Once again, some flippant memory aides can be helpful.

QNH is the *pressure for a No Hit altitude* – it keeps you from hitting the obstacles on your chart.

QFE is the *pressure for Flying an Extra 300* – it helps aerobatic pilots not bust altitude floor.

QNE is the *pressure for Navigating Everywhere* – it is the basis for Flight Levels, and it also gives you the capability to pick one of calibrated airspeed, equivalent airspeed, Mach, or dynamic pressure and predict all of the others.

REFERENCE

Society of Flight Test Engineers (SFTE) Reference Handbook, Third (2013) Edition

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